Multimodal Sensor Fusion for Industrial Machinery Failure Prediction Using Vision Transformers

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ABSTRACT

Relying solely on vibration waveforms often results in delayed warnings for non-rotational industrial faults. We propose ThermoVib-ViT, a multimodal cross-attention Vision Transformer that combines synchronized thermal imaging with tri-axial vibration spectrograms. Across 500 hours of accelerated life testing on industrial gearboxes, ThermoVib-ViT achieved a 96.4% F1-score, predicting catastrophic bearing seizure 6 hours earlier than vibration-only baselines. However, the model requires 45W continuous power on an Nvidia Jetson Orin, restricting practical deployment on autonomous battery-operated nodes.

1. Introduction

Complex manufacturing machinery undergoes degradation through both mechanical wear and thermal distress. Prior literature predominantly evaluates single-sensor configurations. We demonstrate that thermal imagery provides direct spatial heat dissipation maps that give earlier indicators of lubrication starvation than acoustic or vibration sensors alone.

2. Multimodal Sensor Integration

Our experimental setup integrates a FLIR Lepton 3.5 radiometric thermal sensor (160x120 resolution) with a high-frequency piezoelectric accelerometer. We employ patch-based cross-attention where visual thermal tokens modulate vibrational frequency embeddings.

3. Experimental Results

Our thermal imaging plus vibration transformer achieved a 96.4% F1-score across 500 hours of run-to-failure testing. We demonstrate that optical thermal signatures detect impending friction anomalies 6 hours earlier than vibration sensors alone. Single-modality vibration baselines only flagged anomalies 45 minutes before total seizure.

4. Threats to Validity & Limitations

However, our model demands an Nvidia Jetson Orin with 45W continuous power draw, which is entirely infeasible for battery-harvested wireless sensor nodes deployed on rotating factory shafts. Moreover, the optical thermal line-of-sight is easily degraded by oil splashing and heavy dust accumulation in hostile steel mill environments, requiring frequent manual sensor cleaning.

5. Conclusion

Multimodal vision-vibration fusion significantly elevates diagnostic lead time. Addressing edge power consumption and sensor lens fouling represents the critical frontier for real-world plant adoption.